

Reproducibility Guide

Manuscript: TRA-SAE: A Low-Cost Retrieval-Augmented Expert Reasoning System for Mixed Physics-Logic Question Answering with a 4B Language Model

Venue: Expert Systems (Wiley) | **Base model:** Qwen3.5-4B (open weights) | **Hardware:** single NVIDIA A100-SXM4-80GB

This guide documents how to reproduce every table and figure in the manuscript. All systems are evaluated under a single greedy decoding pass (max 1,024 new tokens); no component uses ground-truth labels to decide whether to regenerate a response. Total training cost: US\$18 (full pipeline below US\$64).

1. Environment

Python 3.10+, PyTorch + CUDA, Unsloth/PEFT (LoRA $r=32$, $\alpha=64$), TRL (GRPO), transformers, scikit-learn (TF-IDF router), z3-solver (symbolic verifier), pint (SI-unit normalisation). Install with `pip install -r requirements.txt`. A single A100-80GB (BF16) reproduces all results.

2. Data and the 90/10 split

The EXACT 2026 corpus (1,945 train + held-out eval; 1,213 physics / 732 logic) is combined and divided by a fixed 90/10 stratified split into 1,945 train / 217 validation (141 physics, 76 logic). The exact split is released under `processed_data/exact_train` and `processed_data/exact_val`. **The EXACT competition data are governed by the challenge access policy and must NOT be redistributed** — the repository ships the split indices/IDs and the loader, not the raw licensed content. MMLU and FOLIO are downloaded from their public sources by the eval scripts.

3. Training pipeline (three stages)

Stage	Script	Config	Output adapter	Time/Cost
Phase 1: SFT	<code>run_phase1_sft.py</code>	3 ep, lr 2e-4, eff. batch 16	<code>checkpoints/phase1_sft</code>	58.8 min / \$3.60
Phase 1.5: Logic SFT	<code>run_phase1_5_logic_sft.py</code>	2 ep, lr 5e-5, logic only	<code>checkpoints/qwen35_sft_logic</code>	15.7 min / \$0.96
Phase 2: GRPO	<code>run_phase2_grpo.py</code>	250 steps, K=4, beta 0.04, lr 1e-6	<code>checkpoints/qwen35_grpo</code>	79.8 min / \$4.89

Reward (Phase 2): $R = 0.30 r_{\text{format}} + 0.60 r_{\text{correct}} + 0.10 r_{\text{unit}} - 0.10 [\text{len} > 800 \text{ tok}]$. Correctness uses $\leq 5\%$ relative tolerance after SI-unit normalisation (physics) and normalised string / Z3 matching (logic).

4. Evaluation and configurations

Run `experiments/step0_canonical_eval.py` and `step1_rerun_cfg0_3.py` to produce the per-sample predictions for:

- **cfg0** — Zero-shot (no retrieval, no FT)
- **cfg0-R** — Retrieval only (base + 3 exemplars, no FT)
- **cfg1** — + SFT
- **cfg2** — + Logic SFT
- **cfg3** — + GRPO (best, 47.47%)
- **cfg4** — Dual-LoRA + TF-IDF router
- **cfg5** — Full agent, self-consistency x5

Baselines (best-of-three prompt: XML / boxed CoT / plain CoT): `experiments/step8_fair_baselines.py` (and `step3_baselines.py`). External transfer (MMLU physics, FOLIO logic): `experiments/step9_external_benchmark.py`. Tool-augmented (Python + Z3): `step13_tool_baselines.py`. Reward ablation: `step5_reward_ablation.py`. Statistics (McNemar, Wilson): `step4_stats_and_errors.py`, `step7_stats_extra.py`. Leakage audit: `step10_leakage_check.py`. Multi-seed: `step2_multiseed_cfg3.py` (seeds 42/1337/2024).

5. Table / figure -> artifact map

See **Result_Artifacts_Summary.xlsx** (and .pdf) for the exact script and raw JSON log behind every table and figure.

6. Repository layout

run_phase*.py (training) · experiments/step*.py (evaluation/analysis) · checkpoints/ (LoRA adapters + router.pkl) · processed_data/ (exact split) · logs/*.json (raw per-sample predictions & cost logs) · paper/ (manuscript).
Recommended license: MIT or Apache-2.0 for code.

7. Note on absolute scores

The contribution is a controlled attribution, not a leaderboard result: retrieval (+9.22 pp) and SFT (+5.07 pp) carry the gain, while GRPO, self-consistency and tool augmentation add no statistically separable benefit at the 217-sample validation size. Logic remains the principal bottleneck (<23% across configs).